



Hyunwoo Park

AI RESEARCHER, DATA SCIENTIST

405, 10-7, Gwangun-ro 7-gil, Nowon-gu, Seoul, Republic of Korea

☎ (+82) 10-6381-9680 | ✉ bhw119@gmail.com | 📱 bhw119 | 🎓 Google Scholar

"Precision in every data point."

Summary

M.S. student in Applied Artificial Intelligence at Kwangwoon University, advised by Prof. Minsu Cho. My research centers on **label-efficient learning**, i.e., how models can generalize from scarce supervision, with applications in **remote sensing time-series**, **process mining**, and **NLP**. I am also interested in when modern sequence architectures (attention, Transformers) actually pay off outside the large-data regime they were designed for.

Kwangwoon University

Seoul, S.Korea

M.S. IN APPLIED ARTIFICIAL INTELLIGENCE

Mar. 2026 - Feb. 2028 (Expected)

- Department of Applied Artificial Intelligence, advised by **Prof. Minsu Cho** (DA. Lab)
- Research focus: Label-efficient learning (few-shot learning for satellite time-series, federated learning for predictive process monitoring)

Kwangwoon University

Seoul, S.Korea

B.S. IN INFORMATION CONVERGENCE, DATA SCIENCE

Mar. 2020 - Feb. 2026

- **Major GPA: 4.31 / 4.5** (Cumulative GPA: 4.13 / 4.5)

Research Experience

PADA LAB (Process Automation & Data Analytics Lab)

Seoul, S.Korea

GRADUATE RESEARCHER (M.S. STUDENT)

Mar. 2026 - Present

- Researching **few-shot learning for satellite time-series**, classifying IUCN Global Ecosystem Typology (GET) classes from **HLS (Harmonized Landsat-Sentinel-2)** data with only a handful of labels per class.
- Researching **federated learning for predictive process monitoring** on emergency department event logs (MIMICEL), without centralizing patient data.
- Continuing research on dual-process modeling of online review helpfulness (see Publications).

PADA LAB (Process Automation & Data Analytics Lab)

Seoul, S.Korea

UNDERGRADUATE RESEARCH INTERN

Sep. 2024 - Feb. 2026

- Conducted research on the cognitive mechanisms of online review helpfulness across diverse online markets (Amazon, Booking.com, Audible, Coursera) using **400K+ reviews**.
- Proposed a predictive framework based on **Dual-Process Theory**, distinguishing between Systematic (e.g., depth, breadth) and Heuristic (e.g., rating deviation, recency) processing cues.
- Implemented NLP pipelines to extract features, utilizing **RoBERTa**-based representations for emotional arousal analysis and **NMF (Non-negative Matrix Factorization)** for measuring topic breadth.
- Applied **PLS-SEM** for latent construct validation and developed a **Multilevel ZINB (Zero-Inflated Negative Binomial)** model to robustly handle zero-inflated count data and item-level heterogeneity.
- Identified key findings: (1) Systematic cues consistently drive helpfulness across all domains, whereas (2) Heuristic cues exert stronger influence in experience-goods platforms (e.g., Audible).
- Resulted in a co-authored paper in *IEEE Access* (SCIE, 2026) and a first-author paper under review at *Electronic Commerce Research* (Springer).

Publications

Information Processing and Review Helpfulness in Online Platforms: A Dual-Process Perspective

IEEE Access (SCIE)

CO-AUTHOR (PUBLISHED)

2026

Heeryeong Park, Pilkyu Chung, **Hyunwoo Park**, Sujin Jung, Minsu Cho

When Does Systematic Processing Add Value? A Dual-Process Analysis of Review Helpfulness across Online Markets

Electronic Commerce Research (Springer)

FIRST-AUTHOR (UNDER REVIEW)

Hyunwoo Park, Sujin Jung, Heeryeong Park, Minsu Cho

Design and Implementation of ‘Lec-Q’: A Two-Way Learning Support Service to Enhance Understanding of Real-Time Online Lectures

FIRST-AUTHOR (PUBLISHED)

Hyunwoo Park, SeongYun Cheon, AhReum Yoo, Kyudong Park

Proceedings of HCI Korea 2026

2026

Proposal of Neuro-Symbolic AI-based Dark Pattern Detection and User Assistance System in Elderly Kiosk Environments

CO-AUTHOR (PUBLISHED)

Sujin Jung, Soyoung Park, Hyunwoo Park, MinJin Oh, Minsu Cho

ICROS 2025 (Institute of Control, Robotics and Systems Annual Conference), pp. 958-959

2025

Development of ‘PHOTATO’, a Photo Analysis Travel Recommendation System

CO-AUTHOR (PUBLISHED)

HaKyeong Lee, HanBee Ko, EunJin Joo, Hyunwoo Park, SeongYun Cheon, SeoYeon Jeon, YongMin Kim, Wonjoon Kim

Proceedings of HCI Korea 2025, pp. 1211-1215

2025

Projects

Few-Shot Learning for Ecosystem Classification from Satellite Time-Series

Seoul, S.Korea

GRADUATE RESEARCH

Mar. 2026 - Present

- **Goal:** Classify IUCN Global Ecosystem Typology (GET) classes from satellite time-series with only a handful of labeled samples per class.
- **Data:** HLS (Harmonized Landsat-Sentinel-2) time-series over South Korea, covering 24 GET classes across 6 ecological realms.
- **Approach:** ProtoNet variants over InceptionTime / Presto encoders, hierarchical margin losses, and sibling-repulsion regularization for fine-grained, same-realm class separation.

Federated Learning for Predictive Process Monitoring

Seoul, S.Korea

GRADUATE RESEARCH

2026

- **Goal:** Predict ED-to-inpatient admission from emergency department event logs without centralizing patient data.
- **Data:** MIMICEL v2.1.0 (MIMIC-IV ED event logs, ~425K cases).
- **Findings:** Compared FedAvg / FedProx / Ditto / SCAFFOLD under realistic label skew across sites; FedProx’s advantage grows monotonically with heterogeneity.

Lec-Q: AI-based Interactive Learning Platform

Seoul, S.Korea

PROJECT LEAD & MAIN DEVELOPER (FIRST AUTHOR)

Sep. 2025 - Dec. 2025

- Developed a real-time interactive educational platform using WebRTC to resolve communication gaps in online lectures.
- Implemented a RAG (Retrieval-Augmented Generation) based AI Q&A chatbot (OpenAI API) that answers from lecture materials, featuring a 2-step verification by instructors.
- Designed real-time handwriting sharing that automatically captures and distributes the instructor’s notes to reduce students’ cognitive load.
- Built an analytics dashboard for instructors, visualizing time-stamped question distribution and student concentration levels.
- Stack: React, Node.js, WebRTC, OpenAI API. Published as First Author at HCI Korea 2026.

Neuro-Symbolic AI-based Dark Pattern Detection Service

Seoul, S.Korea

FULL STACK DEVELOPER & AI RESEARCHER (GRADUATION PROJECT)

Jan. 2025 - Sep. 2025

- Detected deceptive designs (Dark Patterns) on websites and kiosks using Neuro-Symbolic AI, combining BERT-based deep learning with logical reasoning for high precision.
- Delivered a Chrome Extension & Web Dashboard; led full-stack development using React, Node.js, and Flask, handling AI model serving and MongoDB integration.
- Expanded the research to offline environments (Kiosk) to assist elderly users, focusing on Human-Robot Interaction (HRI).
- Published a paper at ICROS 2025 (HRI Field).

PHOTATO: Photo Analysis-based Personalized Travel Recommendation System

Seoul, S.Korea

TEAM LEAD & BACKEND DEVELOPER (CO-AUTHOR)

Nov. 2024 - Feb. 2025

- Led the development of a travel recommendation service that suggests destinations based on visual features of user-uploaded photos using CNN (Convolutional Neural Networks).
- Engineered the backend logic for route optimization by applying the TSP (Traveling Salesperson Problem) algorithm to generate efficient travel paths.
- Conducted comprehensive user research (Surveys, IDIs) and derived UX insights using Personas and Journey Maps to define core features.
- Published the paper in Proceedings of HCI Korea 2025.

Honors & Awards

DOMESTIC & ACADEMIC

Sep. 2025 **Encouragement Award (3rd Place)**, Graduation Project Exhibition
 Mar. 2025 **Academic Excellence Scholarship**, Kwangwoon University
 Dec. 2024 **Grand Prize (Poster Session)**, MATCH-UP Competition
 Dec. 2024 **Grand Prize (Practical Report)**, MATCH-UP Competition
 Sep. 2024 **Academic Excellence Scholarship**, Kwangwoon University
 Sep. 2024 **Full Tuition Scholarship (1st class)**, Kwangwoon University
 Sep. 2024 **Dean's List**, Kwangwoon University
 Sep. 2023 **Academic Excellence Scholarship**, Kwangwoon University

Seoul, S.Korea
 Seoul, S.Korea
 Seoul, S.Korea
 Seoul, S.Korea
 Seoul, S.Korea
 Seoul, S.Korea
 Seoul, S.Korea

Teaching Experience

D.Lab Code Academy

Seoul, S.Korea

CODING INSTRUCTOR

Dec. 2024 - May 2025

- Instructed elementary school students in fundamental coding concepts and computational thinking.
- Facilitated hands-on programming activities to foster problem-solving skills and logical reasoning for young learners.

Kwangwoon Artificial Intelligence High School

Seoul, S.Korea

LEAD INSTRUCTOR & CURRICULUM DEVELOPER

Jan. 2024 - Nov. 2024

- Collaborated with school teachers starting in January to design the syllabus and create teaching materials for the Python special course.
- Conducted the 1st semester course (10 sessions, Mar. 07 - Jul. 12) covering Python fundamentals (Control Flow, Functions, Data Structures) and Data Analysis libraries (Matplotlib, Numpy, Pandas).
- Selected as the instructor for the 2nd semester continuously in recognition of excellent teaching evaluations and student feedback.
- Conducted the 2nd semester course (10 sessions, Sep. 25 - Nov. 29) focusing on Web Crawling, Big Data Analysis (Text mining, Geo-spatial, Sentiment Analysis, Recommendation Systems), and supervising final team projects.

Extracurricular Activity

CHIC (Data Science & AI Academic Club)

Seoul, S.Korea

CORE MEMBER & ACADEMIC MENTOR

Sep. 2023 - Feb. 2026

- Provided academic mentoring for 2nd and 3rd-year Data Science majors, focusing on core coursework and problem-solving (Mar. 2025 - Jun. 2025).
- Conducted intensive Deep Learning & AI studies, covering theoretical foundations and model architectures (Dec. 2023 - Feb. 2024).
- Participated in regular academic sessions on Python programming, data analysis, and machine learning algorithms.

Sowon (SW Volunteer Corps)

Seoul, S.Korea

TEAM LEADER OF CURRICULUM DIV. & LEAD INSTRUCTOR

Sep. 2023 - Feb. 2025

- Served as the Lead Instructor at Kwangwoon Artificial Intelligence High School, teaching AI and software concepts for one year.
- Participated as a Mentor for the 'And Center' Youth Hackathon to guide student projects (Nov. 2023, Nov. 2024).
- Conducted coding education volunteering at Seoul Jeongmin Special School (Dec. 2023 - Mar. 2024).

Student Council

Seoul, S.Korea

STUDENT COUNCIL MEMBER & DEPARTMENT REPRESENTATIVE

Mar. 2024 - Feb. 2026

- Represented the department at the Freshman Orientation, delivering a presentation on the academic curriculum and career paths (Feb. 2025).
- Provided consultation and Q&A sessions for undeclared majors regarding major selection and educational roadmap.